

BIOSTAT724 Homework 1

Note: Only use R (or other software) if it is indicated in the question. Full credit will not be given to answers without work shown.

1. (15 pts) In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and $1 - p$ the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question, given that he or she answered it correctly? What happens when m is very large? What happens when p is close to 0?
2. (20 pts) Suppose urn H is filled with 40% green balls and 60% red balls, and urn T is filled with 60% green balls and 40% red balls. Someone will flip a coin and then select a ball from urn H or urn T depending on whether the coin lands heads or tails, respectively. Let X be 1 or 0 if the coin lands heads or tails, and let Y be 1 or 0 if the ball is green or red.
 - (a) Write out the joint distribution of X and Y in a table.
 - (b) Find $E(Y)$. What is the probability that the ball is green?

- (c) Find $\text{Var}(Y|X=0)$, $\text{Var}(Y|X=1)$ and $\text{Var}(Y)$. Thinking of variance as measuring uncertainty, explain intuitively why one of these variances is larger than the others.
- (d) Suppose you see that the ball is green. What is the probability that the coin turned up tails?

3. (10 pts) Let θ be a $\text{Beta}(a,b)$ random variable, and suppose that the conditional distribution of X , given that θ , is $\text{Binomial}(n, \theta)$. Find the (marginal) probability mass function of X .

4. (20 pts) Full conditionals: Let X, Y, Z be random variables with joint density (discrete or continuous) $p(x, y, z) \propto f(x, z)g(y, z)h(z)$. Show that

- (a) $p(x|y, z) \propto f(x, z)$, i. e., $p(x|y, z)$ is a function of x and z ;
- (b) $p(y|x, z) \propto g(y, z)$, i. e., $p(y|x, z)$ is a function of y and z ;
- (c) X and Y are conditionally independent given Z .

5. (20 pts) PH 3.1

6. (15 pts) PH 3.2